

Kartikeya Vemula

Data Analyst | Product Analytics | Business Intelligence | Analytics Engineering
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Profile

Data Analyst with experience in SQL, Python, ETL pipelines, business intelligence, customer analytics, KPI monitoring, and data warehousing. Built analytics systems across retail, product, operations, and engineering datasets, translating raw data into executive dashboards, retention insights, funnel diagnostics, and decision-ready business reporting.

Core Skills

Analytics:	SQL, Data Analytics, Business Analysis, Business Analytics, Business Intelligence, Product Analytics, Customer Analytics, KPI Monitoring, Key Performance Metrics (KPIs), Performance Monitoring, Funnel Analysis, A/B Testing, Cohort Analysis, CLV, Retention Analytics, Root Cause Analysis
Data Engineering:	ETL Pipelines, Data Warehousing, Data Modeling, Star Schema, Data Ingestion, PostgreSQL, BigQuery, REST APIs, Data Quality, Quality Assurance
Tools:	Python, Excel, Power BI, Tableau, Plotly, Looker Studio, Business Intelligence Tools, Data Visualization, Git, Docker, VS Code
Statistics & ML:	Statistical Analysis, Applied Mathematics, Predictive Analytics, Anomaly Detection, Artificial Intelligence (AI), Machine Learning, Forecasting, Time Series Analysis, XGBoost, SHAP
Professional:	Stakeholder Communication, Communication Skills, Cross-Functional Collaboration, Executive Reporting, Decision Making, Problem Solving, Attention to Detail, Process Improvement

Projects

Customer Segmentation & Retention Intelligence Platform Python, SQL, PostgreSQL, RFM, CLV, Plotly

- Built an RFM segmentation engine on 1M+ retail transactions, classifying 5,876 customers into 8 behavioral tiers using SQL window functions and customer analytics logic; Champions (22.1% of customers) drove 68.35% of total revenue — quantifying business performance and retention ROI.
- Identified £171K+ revenue exposure across 15 high-value inactive customers; delivered cohort retention matrices, CLV analysis, churn-risk dashboards, and executive reporting to support lifecycle marketing decisions.

Instacart Customer & Operations Intelligence Platform Python, PostgreSQL, Docker, SQL, Apriori, Plotly Dash

- Engineered a Dockerized PostgreSQL warehouse ingesting 3.4M+ orders across 206K customers and 49K SKUs for customer behavior analysis, merchandising analytics, market research, reorder analytics, and business performance reporting.
- Built RFM segmentation identifying 55,092 at-risk customers (26.7% of base) as primary revenue recovery opportunity; executed market basket analysis via Apriori with 3.26 lift and 29.1% confidence on top product association rules.
- Built a 4-tab Plotly Dash application (Executive Overview, Customer Lifecycle, Merchandising, Market Basket) connected live to PostgreSQL, surfacing reorder rates, cohort decay, and SKU velocity.

Customer Intelligence Data Warehouse Python, PostgreSQL, Docker, SQL, ETL, Star Schema, Power BI

- Ingested, validated, and cleaned 1,067,371 raw e-commerce transactions using data quality and quality assurance controls with strong attention to detail; built SQL analytics layers for AOV, repeat purchase rate, LTV, and geographic BI across 43 countries.
- Developed executive data visualization dashboards in Power BI and Looker Studio for business intelligence reporting and performance monitoring.

KPI Reliability & Diagnostic Engine Python, SQL, BigQuery-Compatible Schema, Plotly, YAML

- Built an end-to-end KPI monitoring pipeline with 0–100 data health gates, volume checks, null checks, anomaly detection, and structured JSON diagnostics.
- Applied analytical problem-solving techniques to automate root cause analysis for KPI drops, performance monitoring, and KPI diagnostics; ranked Mobile/APAC/Safari as primary offender with –9.2% segment-level change.
- Reduced manual analyst investigation effort by converting KPI incidents into dashboard-ready diagnostic summaries and performance monitoring workflows.

Engineering Analytics Data Warehouse & Insight Pipeline Python, GitHub REST API, SQL, ETL, PostgreSQL, Grafana

- Designed an ETL pipeline ingesting 5,000+ commits across 50+ repositories into a star schema with idempotent upserts, pagination handling, and rate-limit-aware API logic.
- Delivered SQL window-function analytics for cumulative commits, monthly velocity, repository activity, and engineering productivity reporting.

Education

M.S. Data Science, University of Maryland Baltimore County, USA GPA: 3.78/4.0 Focus: Machine Learning, Statistical Modeling, Experimental Design, Time Series Analysis	2024–2026
B.Tech Computer Science, GITAM University, India GPA: 8.89/10 First-Class with Distinction	2020–2024